GPT-6.1 SOL

10MAA

End of Year Revision Examination 2026

Student's full name: _____

Form class: _____

Maths teacher's code: _____

Time allowed: 2 hours

Instructions:

- Write in blue/black pen only. Use a pencil for diagrams.
- No twink or correction fluid allowed.
- Calculators are permitted.
- Answer **all questions in this booklet**, showing working in the appropriate space.
- Unless otherwise stated, give non-exact answers to **three significant figures**, angles to **one decimal place**, and money to **two decimal places**.

Topic	**Mark**
Algebraic Expressions & Equations	/34
Statistics & Probability	/26
Trigonometry	/12
Measurement	/14
Coordinate Geometry	/11
Graphs of Quadratic Functions	/24
Geometry	/14
Total	**/135**

Independent revision paper

OpenAI

Useful Formulae

The Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Volume of a cylinder

$$V = \pi r^2 h$$

Volume of a sphere

$$V = \frac{4}{3}\pi r^3$$

The Cosine Rule

$$a^2 = b^2 + c^2 - 2bc\cos A$$

The Sine Rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

The Area of a Triangle

$$A = \frac{1}{2}ab\sin C$$

The midpoint of a line segment

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Distance

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Gradient of a line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Equation of a line

$$y - y_1 = m(x - x_1)$$

Perpendicular criteria

$$m_1 m_2 = -1$$

For lines with defined gradients.

Algebraic Expressions & Equations — 34 marks

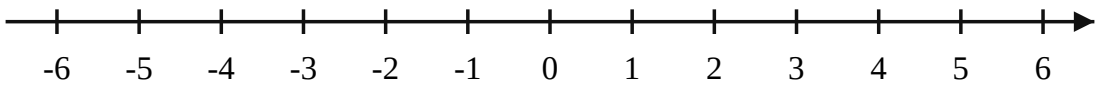
1. Solve $4(2x - 3) = 3x + 23$.

x: ..... [2]

2. (a) Solve the inequality $9 - 4t \leq 3t + 30$.

Answer: ..... [3]

(b) Represent your solution to part (a) on the number line.



[1]

3. Find the value of $3a^2 - 2bc$ when $a = -3$, $b = -2$ and $c = 5$.

Answer: ..... [2]

4. Simplify fully. Assume all variables in this question are positive.

(a)

$$\frac{18p^7q^3}{6p^2q}$$

Answer: ..... [2]

(b)

$$\left(81k^8\right)^{\frac{3}{4}}$$

Answer: ..... [2]

5. (a) Factorise completely $6x^2 - 54$.

Answer: ..... [2]

(b) Express $\frac{5}{x-2} - \frac{2}{x+1}$ as a single fraction in its simplest form.

Answer: ..... [3]

Algebraic Expressions & Equations continued

6. Make r the subject of $T = \sqrt{5r + 4}$.

r: ..... [2]

7. An AI service charges a fixed monthly fee of \$$a$ and \$$b$ per batch of requests.

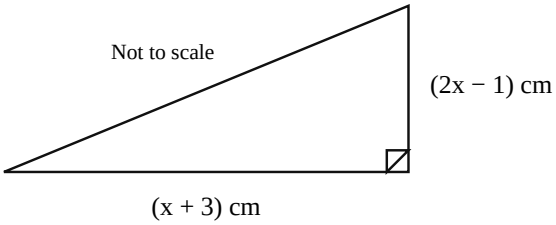
One month, 40 batches cost \$86. Another month, 70 batches cost \$122.

Find the cost of 100 batches in one month.

Cost: ..... dollars [3]

8. The diagram shows a right-angled triangular support for a computer stand. Its area is 30cm^2.

Find the value of x.



x: ..... cm [4]

9. In a fictional benchmark, a Windows computer completes x identical jobs each hour. A Linux computer completes $(x + 10)$ jobs each hour.

   Both computers work at constant rates.

   (a) Write an expression for the time, in minutes, that Windows takes to complete one job.

   Answer: ..... minutes [1]

   (b) Write an expression for the time, in minutes, that Linux takes to complete one job.

   Answer: ..... minutes [1]

   (c) Linux takes 1 minute less per job than Windows. Form an equation in x.

   Answer: ..... [2]

   (d) Solve your equation to find the number of jobs Windows completes each hour.

   Answer: ..... jobs per hour [4]

Statistics & Probability — 26 marks

1. The table shows the number of programming projects completed by 30 students.

Projects	0	1	2	3	4
Number of students	3	6	10	7	4

(a) Find the mode.

Answer: ..... [1]

(b) Find the median.

Answer: ..... [1]

(c) Calculate the mean.

Answer: ..... [2]

2. The table gives results from a fictional test of the same program.

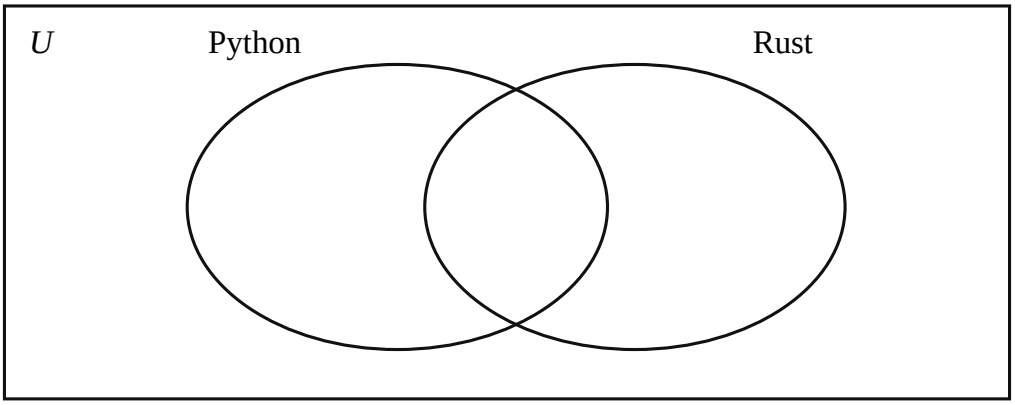
Operating system	Number of runs	Mean run time / seconds
Linux	12	8
macOS	8	11
Windows	5	14

Calculate the mean run time for all 25 runs.

Answer: ..... seconds [3]

3. Of 40 students, 24 use Python, 19 use Rust, and 7 use neither language.

   Complete the Venn diagram and find the number who use exactly one of the two languages.



Answer: ..... students [3]

4. A club tests 100 computers. Each computer either passes or fails a compatibility check.

	Pass	Fail	Total
Linux	36	4	40
macOS	24	6	30
Windows	18	12	30
Total	78	22	100

(a) Find the probability that a randomly selected computer uses Linux and passes.

Answer: ..... [1]

(b) Given that a computer uses macOS, find the probability that it passes.

Answer: ..... [1]

(c) Find the probability that a randomly selected computer uses Linux or fails.

Answer: ..... [1]

(d) Two computers are selected at random without replacement. Find the probability that both are Windows computers that fail.

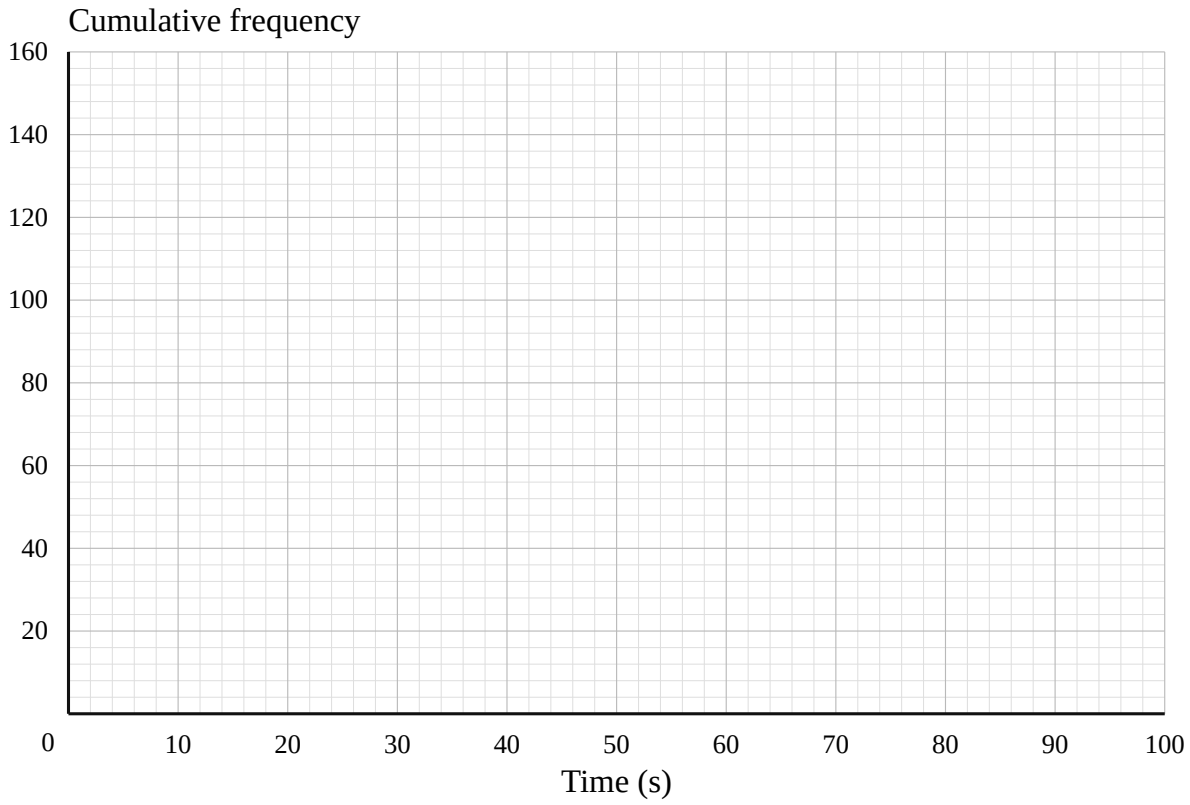
Answer: ..... [2]

5. The table shows the times taken by an AI system to answer 160 prompts.

Time, t / seconds	Frequency	Cumulative frequency
$0 < t \leq 20$	16	
$20 < t \leq 40$	32	
$40 < t \leq 60$	56	
$60 < t \leq 80$	40	
$80 < t \leq 100$	16	

(a) Complete the cumulative frequency column. [1]

(b) Draw a cumulative frequency graph on the grid. [2]



(c) Use your graph to estimate the upper quartile of the response times.

Answer: ..... seconds [1]

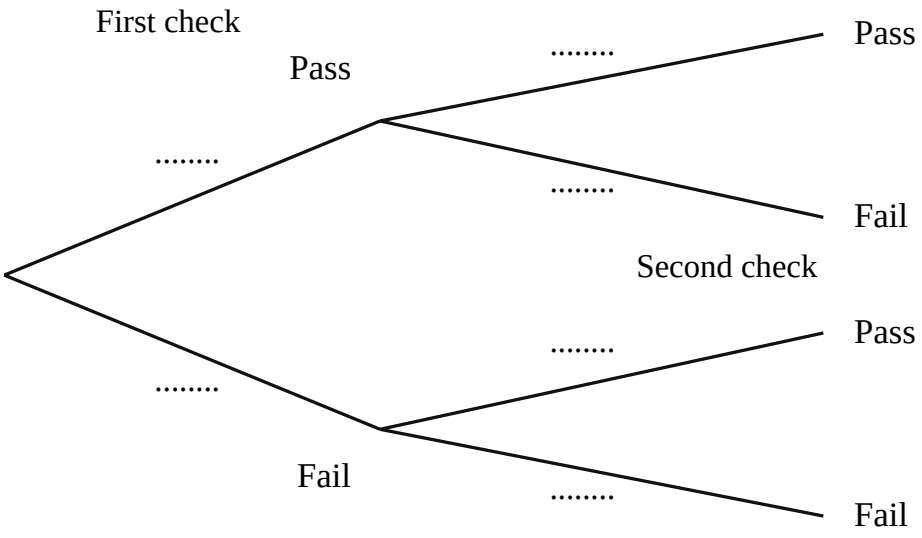
6. AI-generated programs are put through two checks.

The probability that a program passes the first check is $\frac{3}{5}$.

If it passes the first check, the probability that it passes the second is $\frac{3}{4}$.

If it fails the first check, it is repaired. The probability that a repaired program passes the second check is $\frac{1}{2}$.

(a) Complete the probability tree.



[3]

(b) Find the probability that a program passes exactly one check.

Answer: ..... [2]

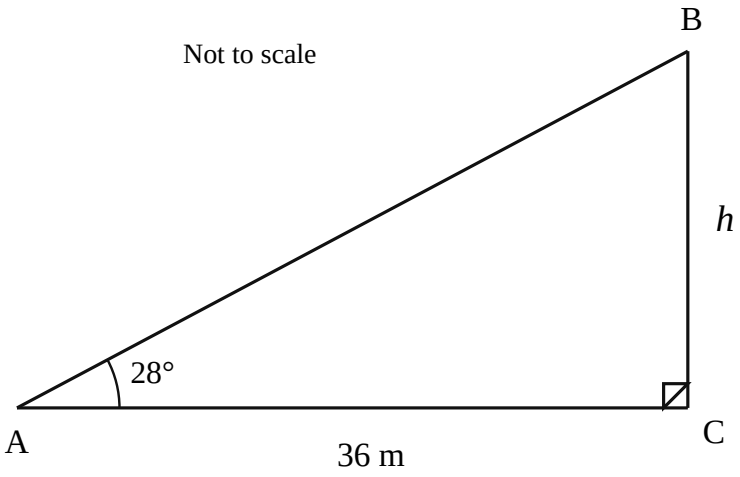
(c) A student claims that more than half of all programs pass the second check. Use probabilities to decide whether the claim is correct.

Answer: ..... [2]

Trigonometry — 12 marks

1. A camera is directly above point C. Point A is 36 m from C on horizontal ground. The angle of elevation of the camera from A is 28°.

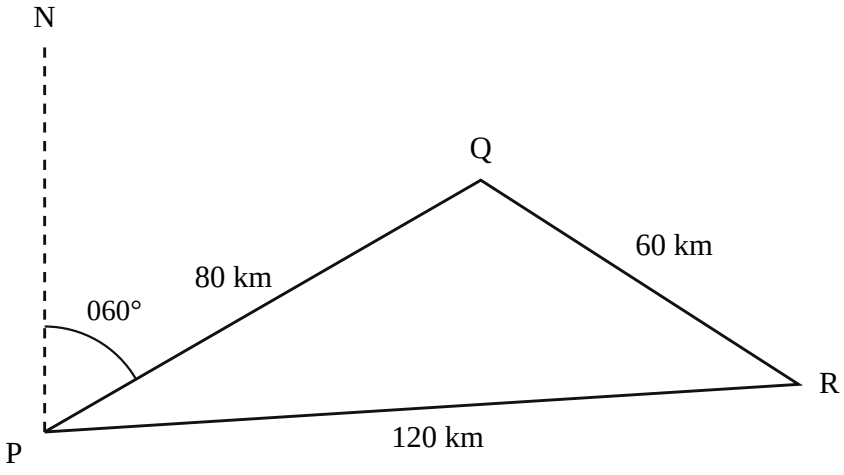
   Calculate the height h of the camera above C.



h: ..... m [2]

2. Three relay stations P, Q and R form a triangle.

   $PQ = 80$ km, $QR = 60$ km and $PR = 120$ km. The bearing of Q from P is 060°. R lies clockwise from Q when viewed from P, as shown.



Not to scale

   (a) Calculate angle QPR.

Answer: ..... ° [2]

2. (b) Find the bearing of:

(i) R from P,

Answer: ..... ° [1]

(ii) P from R.

Answer: ..... ° [1]

(c) Calculate how far R is east of P.

Answer: ..... km [2]

(d) A fourth station S is on the opposite side of QR from P. In triangle QRS, angle RQS is $42°$ and angle QSR is $73°$.

Calculate the distance RS.

RS: ..... km [2]

(e) Calculate the area of triangle PQR.

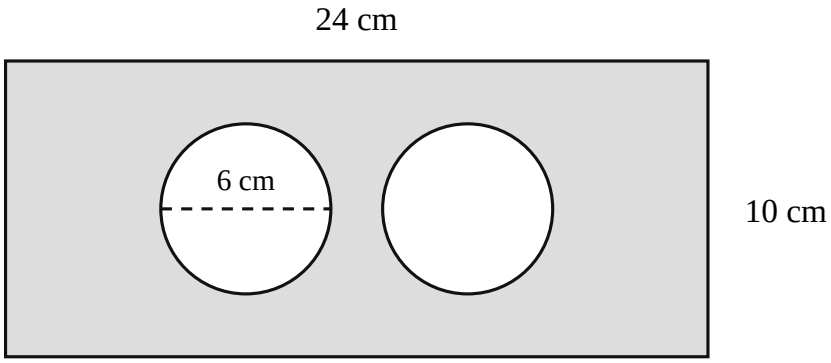
Answer: ..... km^2 [2]

Measurement — 14 marks

1. Convert 0.48m^2 into cm^2.

Answer: ..... cm² [1]

2. A rectangular computer panel is 24 cm long and 10 cm wide. Two circular holes, each of diameter 6 cm, are cut out.

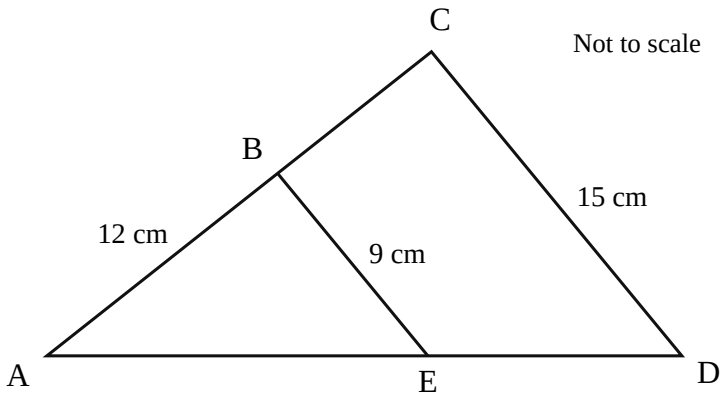


Calculate the area of the remaining panel. Give your answer to the nearest cm^2.

Answer: ..... cm² [3]

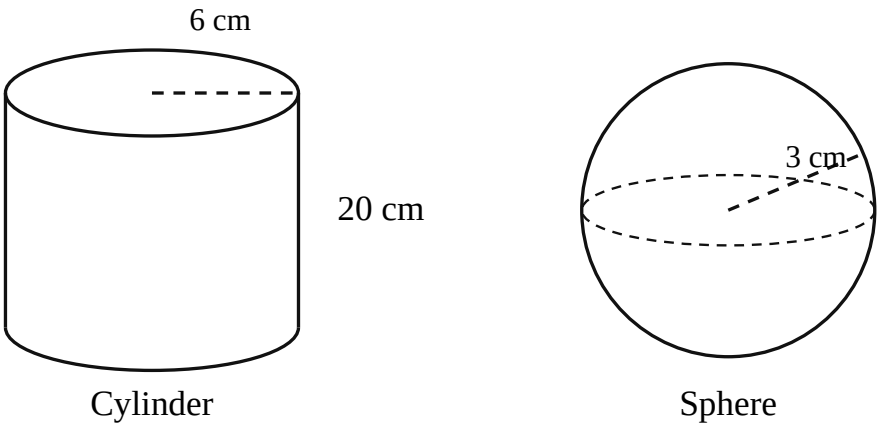
3. In triangle ADC, B lies on AC and E lies on AD, with $BE \parallel CD$.

$AB = 12$ cm, $BE = 9$ cm and $CD = 15$ cm. Calculate BC.



BC: ..... cm [2]

4. A cylindrical container has internal radius 6 cm and height 20 cm. Solid metal spheres each have radius 3 cm.



(a) Calculate the volume of the cylinder.

Answer: ..... cm^3 [1]

(b) Calculate the volume of one sphere.

Answer: ..... cm^3 [1]

(c) The cylinder initially contains $1800 cm^3$ of water. The spheres are lowered into it one at a time and become fully submerged.

Find the greatest number of spheres that can be added without water overflowing. Ignore the volume of any supports.

Answer: ..... spheres [2]

Measurement continued

5. A school installs a straight row of acoustic panels in a computer room.

   The required row length is 24 m, correct to the nearest metre.

   Each panel is 0.80 m long, correct to the nearest centimetre.

   The panels must cover at least the required length. Only complete panels may be bought, and there are no gaps or overlaps. A panel may extend beyond the end of the row.

   Each panel costs $32.50.

   Find the greatest possible difference between the total costs of the panels needed. Show your working clearly.

Difference: ..... dollars [4]

Coordinate Geometry — 11 marks

1. Line L has equation $y = 4 - 2x$.

   (a) State the gradient of L.

   Answer: ..... [1]

   (b) Does the point $(15, -24)$ lie on L? Give a reason.

   Answer: ..... [2]

2. A has coordinates $(-2, 7)$ and B has coordinates $(6, 1)$.

   (a) Find the midpoint of AB.

   Answer: ..... [1]

   (b) Calculate the length of AB.

   Answer: ..... units [2]

   (c) Find the equation of the perpendicular bisector of AB. Give your answer in the form $px + qy = r$, where p, q and r are integers.

   Answer: ..... [5]

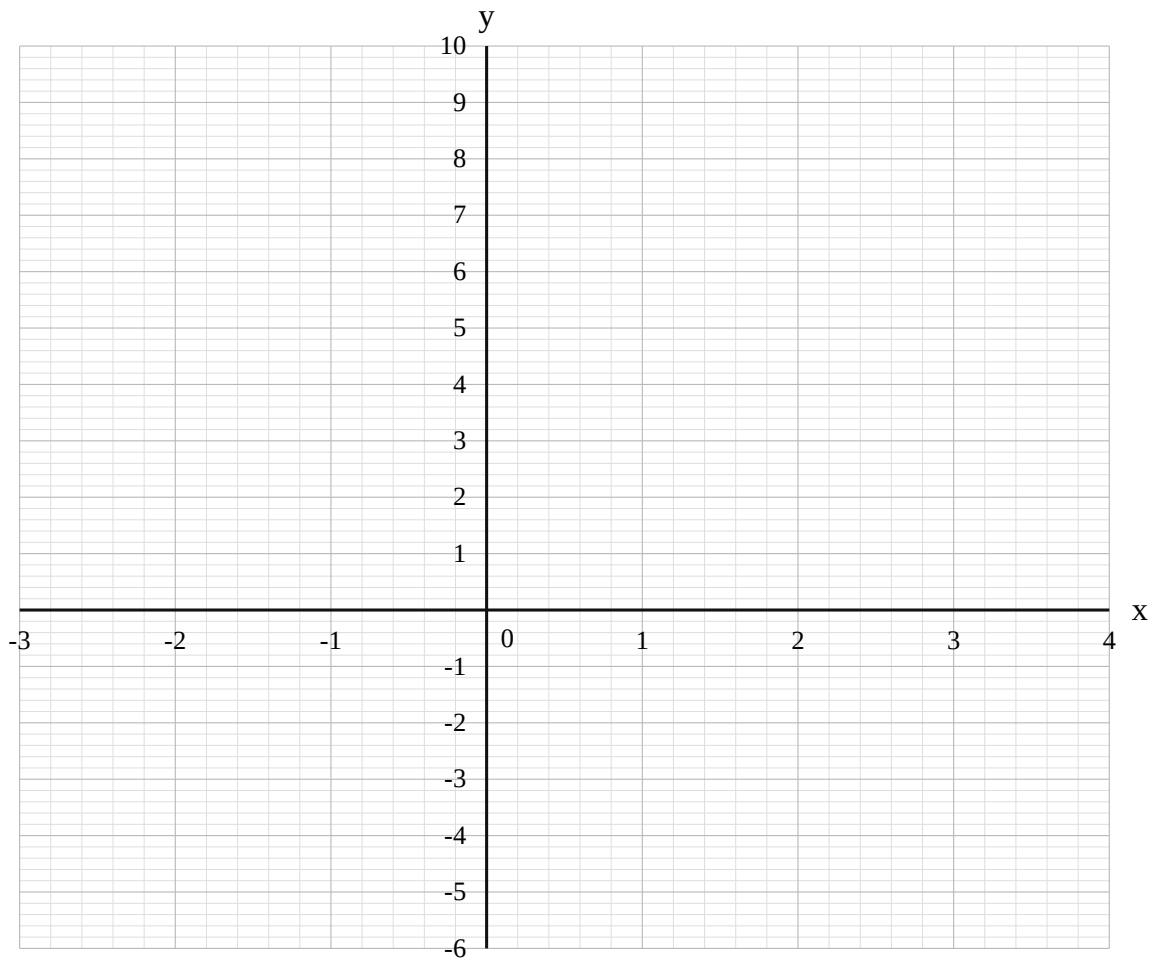
Graphs of Quadratic Functions — 24 marks

1. (a) Complete the table of values for $y = x^2 - x - 4$.

x	−2	−1	0	1	2	3
y						

[2]

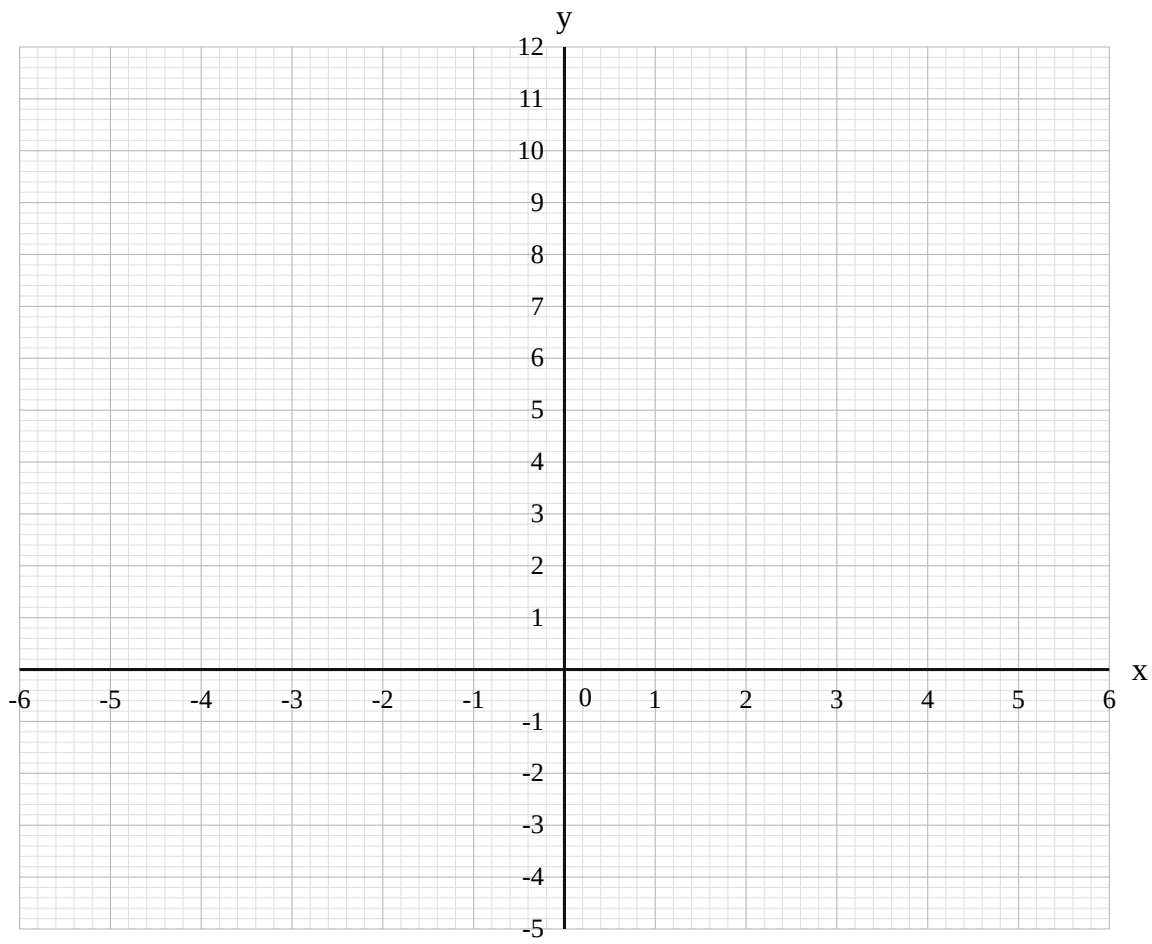
(b) Draw the graph of $y = x^2 - x - 4$ for $-2 \leq x \leq 3$ on the grid.



[3]

2. (a) On the grid, sketch $f(x) = (x + 2)^2 - 4$. Clearly label its vertex and both x-intercepts.

(b) On the same grid, sketch $g(x) = (x - 1)^2 - 1$. Clearly label its vertex. Label each curve with its function name.



(a) [3] (b) [3]

(c) Describe fully the transformation that maps the graph of f onto the graph of g.

[2]

3. A small robot launches a foam ball. Its height h metres above the floor is modelled by

$$h = 1 + 4t - t^2,$$

where t is the time in seconds after launch. The model applies from launch until the ball first reaches the floor.

(a) Find the height at launch.

Answer: ..... m [2]

(b) Calculate the height after 2 seconds.

Answer: ..... m [2]

(c) Find the time when the ball reaches the floor.

Answer: ..... seconds [3]

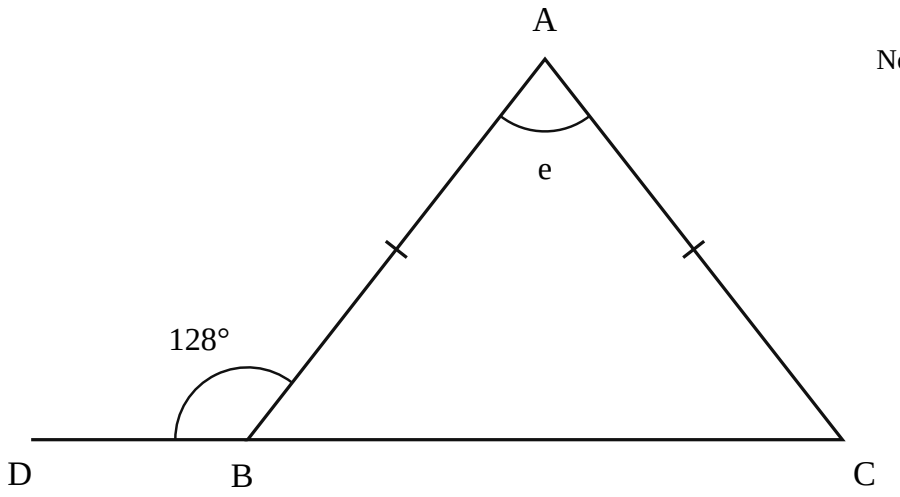
(d) Find all the times, within the stated model, when the ball is 2 m above the floor.

Answer: ..... seconds [4]

Geometry — 14 marks

1. ABC is an isosceles triangle with $AB = AC$. D, B and C lie on a straight line. Angle DBA is 128°.

   Find angle e. Give geometric reasons for each step.



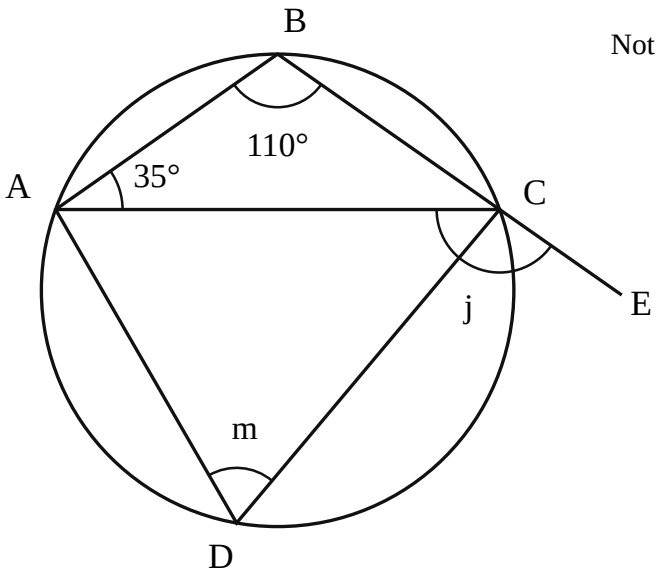
Not to scale

[4]

Geometry continued

2. ABCD is a cyclic quadrilateral. B, C and E lie on a straight line.

   Angle ABC is 110° and angle BAC is 35°.



(a) Calculate angle $m = \angle ADC$. Give a geometric reason.

[2]

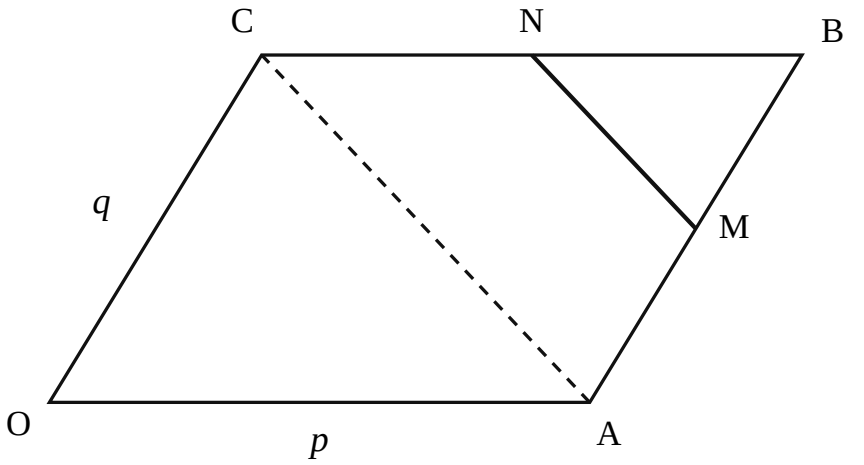
(b) Calculate angle $j = \angle ACE$. Give geometric reasons for your steps.

[4]

Geometry continued

3. OABC is a parallelogram. M is the midpoint of AB, and N is the midpoint of BC.

$\overrightarrow{OA} = \boldsymbol{p}$ and $\overrightarrow{OC} = \boldsymbol{q}$.



Not to scale

(a) Find, in terms of $\boldsymbol{p}$ and $\boldsymbol{q}$:

(i) $\overrightarrow{OM}$,

Answer: ..... [1]

(ii) $\overrightarrow{ON}$.

Answer: ..... [1]

(b) Use a vector method to prove that MN is parallel to AC.

[2]

End of Assessment